APPLIED MACHINE LEARNING AND ECONOMETRICS CASE

Korea Income & Welfare

This project studies how education relates to income in Korean welfare microdata. The updated version keeps the best part of the original delivery: a clear explanation of why this question matters for individuals, educational institutions, organizations, and policymakers.

89,935

Valid records used in the analytical layer

0.645

Best out-of-sample R-squared from the tuned random forest

12.47%

Average income premium per education step in the OLS model

Why this question matters

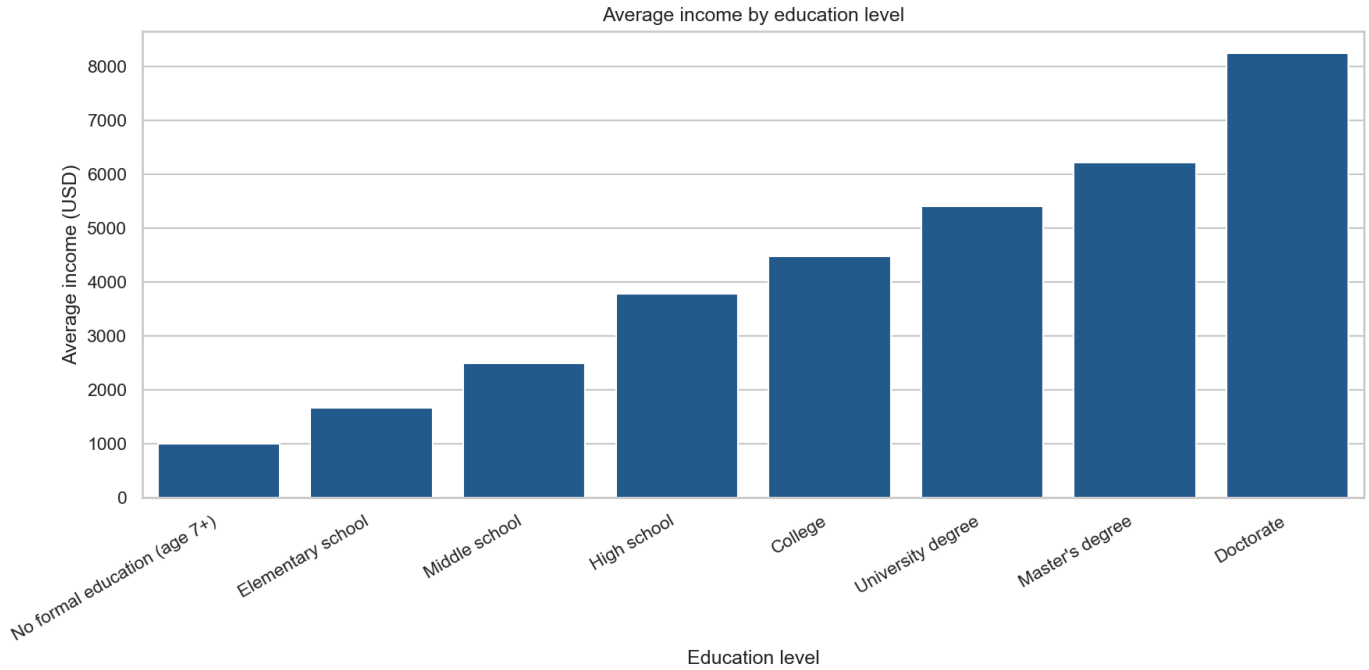
The original notebook framed education and income as a real decision problem. People want to know whether investing in education pays off. Educational institutions want to understand whether credentials are aligned with labor-market value. Policymakers need evidence on how education relates to earnings and mobility. That framing is preserved here because it turns the project into a useful applied case instead of a generic ML exercise.

> Executive takeaway: the project is not just about predicting income. It is about understanding whether education has measurable economic value, how strong that value is, and whether it changes across the income distribution.

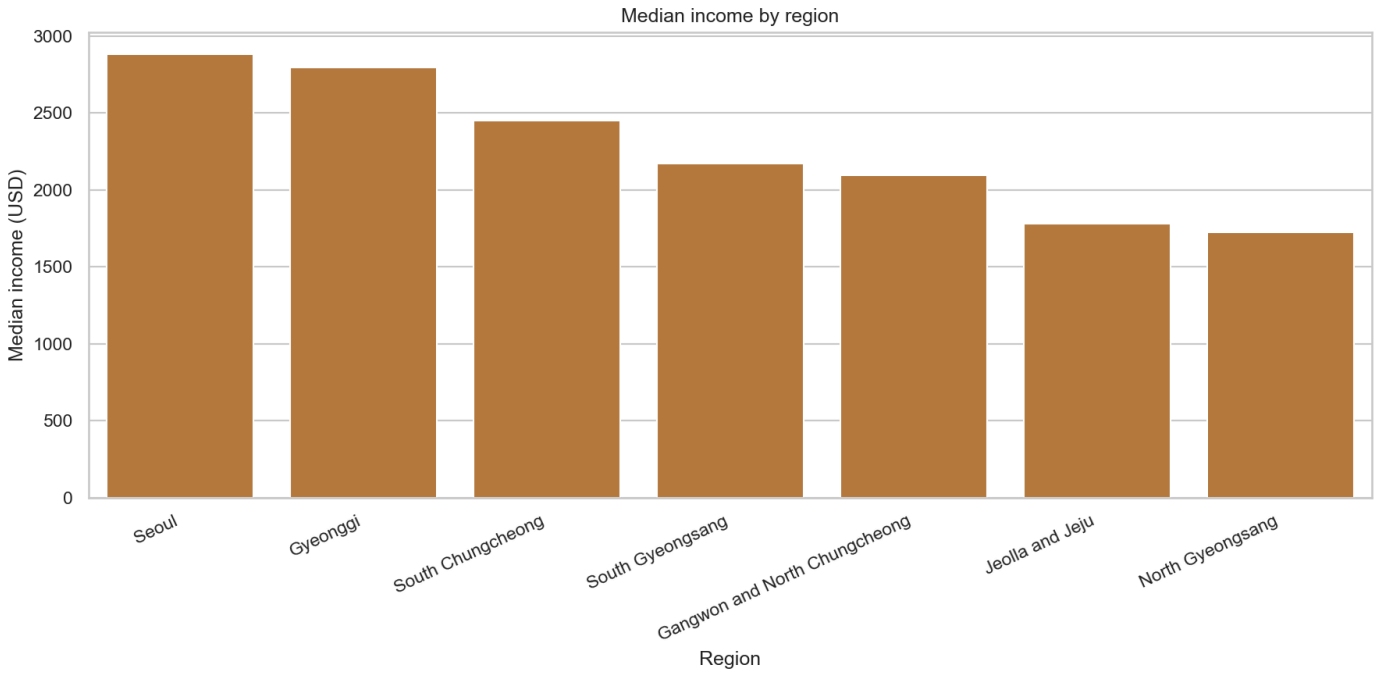
- Does education have a significant effect on income?
- How much does prediction improve once age, family structure, region, and demographics are added?
- Is the payoff to education the same for lower-income and higher-income groups?

What the data show before modeling

The descriptive layer already supports the core intuition: income rises across education levels and varies meaningfully by region. This matters because the strongest analytical projects do not start with the model; they start with a coherent story in the data.



Income increases steadily across the education ladder, which makes the core education-income relationship easy to explain to a non-technical audience.

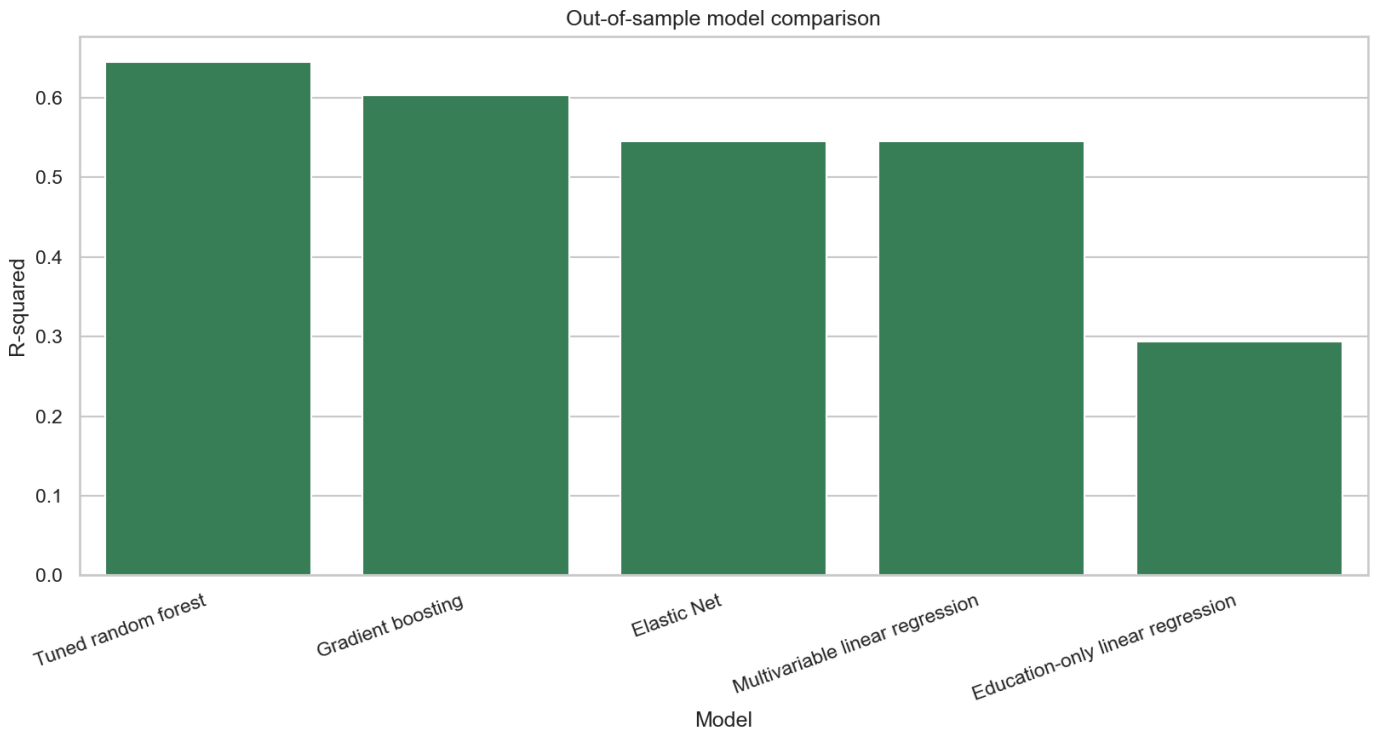


Regional context matters too. Education is important, but it is not the only part of the income story.

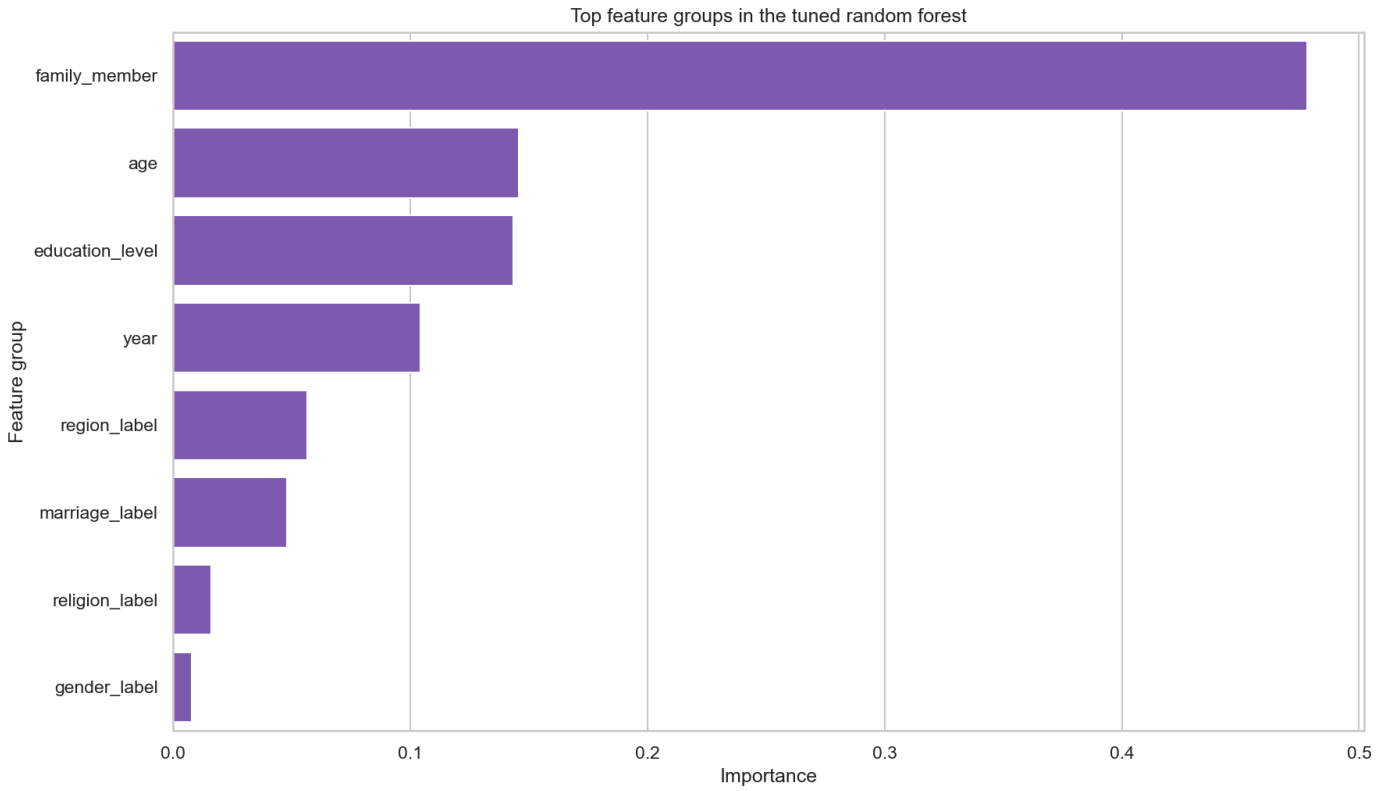
What the modeling layer adds

The repository now compares five out-of-sample predictive models, then complements that with coefficient-based econometrics. This makes the project stronger because it shows both performance and interpretation.

Model	MAE (USD)	RMSE (USD)	R-squared
Tuned Random Forest	880.05	1,249.11	0.645
Gradient Boosting	950.31	1,321.48	0.603
Multivariable Linear Regression	1,049.83	1,414.49	0.545
Education-only Linear Regression	1,345.40	1,762.56	0.294



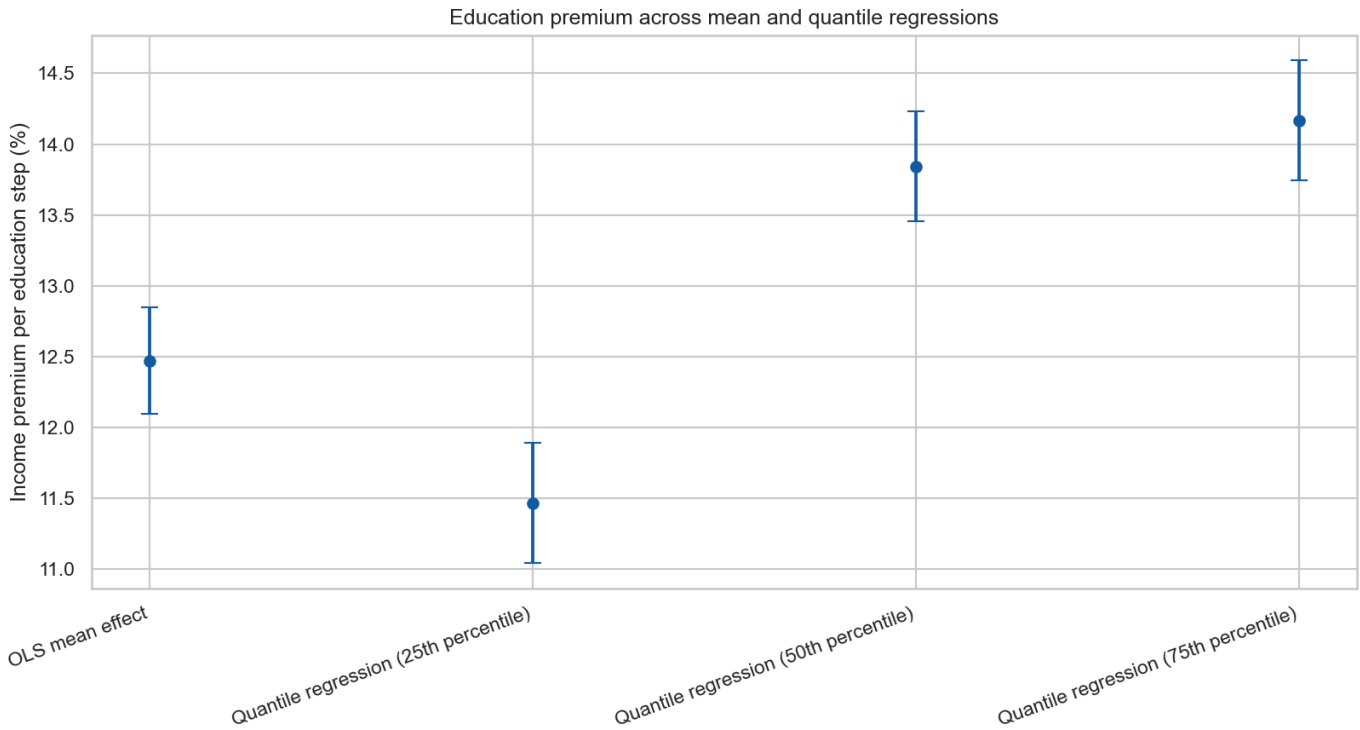
The tuned random forest is the best predictive model. The key point is not only that it wins, but that it quantifies how much value is added by moving from a one-variable benchmark to a richer socioeconomic specification.



Household size, age, education, and survey year are the strongest feature groups in the best model. The result broadens the story without weakening the role of education.

What the econometrics add

The econometric layer gives the project a more interview-friendly conclusion. Instead of only saying that education predicts income, it shows that the education premium itself changes across the income distribution.



The estimated education premium is 12.47% on average, 11.46% near the lower end of the distribution, and 14.17% in the upper part. That makes the story richer than a single mean effect.

Hiring takeaway: this project is strong because it combines hypothesis framing, tuning, benchmarking, and interpretation. It shows not only that performance improved, but also what changed in economic terms.